Functional Annotation Report: PrpF (Q88KF3 / PP_2337)

Gene: prpF · **Locus:** PP_2337 · **UniProt:** Q88KF3 (Q88KF3_PSEPK) **Organism:** *Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) **Family/domains:** PrpF family; Pfam PF04303 (PrpF); InterPro IPR012709 (PrpF), IPR007400 (PrpF-like) **Length:** 396 aa · **KEGG ortholog:** K09788 (2-methylaconitate isomerase, EC 5.3.3.-)

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1. Summary (Answer to the Research Question)

PrpF (PP_2337) is a cofactor-independent 2-methylaconitate (cis–trans) isomerase that functions in the cytoplasm as an essential accessory enzyme of the AcnD-dependent branch of the 2-methylcitric acid cycle (2-MCC), the route by which *Pseudomonas putida* catabolizes propionate. Its specific reaction is the geometric isomerization of the 2-methyl-*trans*-aconitate produced by the Fe/S dehydratase AcnD (PP_2336) into 2-methyl-*cis*-aconitate, the only isomer that the housekeeping aconitase (AcnB) can hydrate to 2-methyl-D-isocitrate. Without PrpF the AcnD dehydration step is a metabolic dead end. This assignment is transferred with high confidence from the experimentally characterized, 83.9%-identical *Shewanella oneidensis* ortholog and is reinforced by the enzyme's location in a complete methylcitrate operon and by perfect conservation of the catalytic residues Lys73 and Cys107.

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2. Gene Identity Verification

The target was rigorously confirmed as the correct protein:

- **Symbol match:** UniProt lists gene name `prpF`, ordered locus `PP_2337`; the submitted protein name "Aconitate isomerase" is consistent with the characterized PrpF function.
- **Family match:** Pfam PF04303 and InterPro IPR012709/IPR007400 all define the **PrpF family**, matching the target's stated domains exactly.

- **Orthology match:** Global Needleman–Wunsch alignment against the reviewed Swiss-Prot entry **Q8EJW4 (PRPF_SHEON, "2-methyl-aconitate isomerase," PE=1)** gives **83.9% identity (328/391 positions)**. The literature I found describes exactly this protein/family — there is **no gene-symbol ambiguity**.
- **Organism match:** UniProt proteome UP000000556, KEGG `ppu:PP_2337`, chromosome-encoded — all *P. putida* KT2440.

The functional evidence below is therefore drawn from direct experimental studies on close PrpF orthologs (*S. oneidensis*, *Salmonella enterica* complementation system) combined with organism-specific bioinformatic/genomic-context evidence for *P. putida*.

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3. Primary Function: The Reaction Catalyzed

3.1 Metabolic problem PrpF solves

Bacteria degrade propionate (and odd-chain fatty-acid–derived propionyl-CoA) via the **2-methylcitric acid cycle**: propionyl-CoA + oxaloacetate → 2-methylcitrate (PrpC) → **2-methyl-*cis*-aconitate** (a dehydration) → 2-methyl-D-isocitrate (aconitase) → pyruvate + succinate (PrpB). The dehydration of 2-methylcitrate is carried out by one of two alternative enzyme systems:

- **PrpD** — a cofactor-less 2-methylcitrate dehydratase (EC 4.2.1.79) that directly yields **2-methyl-*cis*-aconitate**, the aconitase substrate. Organisms using PrpD do **not** need PrpF.
- **AcnD** — an Fe/S, aconitase-like 2-methylcitrate dehydratase (EC 4.2.1.117) that yields a **different geometric isomer** of 2-methylaconitate (2-methyl-*trans*-aconitate), which aconitase **cannot** use. Organisms using AcnD **require PrpF** (Grimek & Escalante-Semerena, *J. Bacteriol.* 2004, P 14702315).

3.2 PrpF's catalytic role

PrpF is the isomerase that reconciles the AcnD product with aconitase:

> "the SoPrpF protein isomerizes the product of the AcnD reaction into the PrpD product (2-MCA], a known substrate of the housekeeping aconitase (AcnB]" — Rocco *et al.*, *J. Bacteriol.* 2017 (P 29145506).

Thus PrpF catalyzes:

2-methyl-*trans*-aconitate ⇌ 2-methyl-*cis*-aconitate (EC 5.3.3.-, KEGG K09788)

Mechanistically it is a **non-PLP, cofactor-less cis–trans isomerase**: a base-catalyzed proton abstraction coupled to rotation about the C2–C3 bond of 2-methylaconitate (Garvey *et al.*, *Protein Sci.* 2007, P 17567742).

On the unmethylated analog PrpF interconverts **trans-aconitate ⇌ cis-aconitate** in vitro, the assay used to demonstrate its activity:

> "Results from in vitro studies show that PrpF isomerizes trans-aconitate to cis-aconitate" (P 17567742).

Careful in-vitro/NMR work refined the model: AcnD dehydrates 2-methylcitrate to an isomer of 2-methyl-*cis*-aconitate (proposed to be 4-methylaconitate / 2-methyl-*trans*-aconitate), which is not an aconitase substrate; PrpF converts it to the productive isomer (P 29145506; P 14702315).

3.3 Substrate specificity

The physiological substrate is the **C7 branched tricarboxylate 2-methylaconitate** (both geometric isomers). PrpF also acts on the non-methylated **aconitate** (citrate-derived) as a model substrate, consistent with a tricarboxylic-acid isomerase active site. There is **no aconitase (hydratase) activity** — PrpF only isomerizes; it does not add/remove water:

> "No aconitase-like activity was found for PrpF" (P 14702315).

3.4 Catalytic residues (structural/bioinformatic evidence)

The *S. oneidensis* PrpF crystal structure (2.0 Å, with bound trans-aconitate; and 1.22 Å K73E variant with trans-aconitate/malonate) shows a **two-domain, pseudo-2-fold fold of the PLP-independent isomerase superfamily** — closely resembling diaminopimelate epimerase and proline racemase (P 17567742 ; P 29145506). Catalysis uses a pair of general acid/base residues; **Lys73** and **Cys107** are essential:

> "SoPrpF variants with substitutions of residues K73 or C107 failed to support growth with propionate" (P 29145506).

In my alignment, **both Lys73 and Cys107 are perfectly conserved in the *P. putida* ortholog** (identical alignment columns), providing strong bioinformatic support that Q88KF3 uses the same catalytic machinery and reaction.

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4. Biological Process & Pathway Context

PrpF operates within the **2-methylcitric acid cycle for propionate degradation**. In *P. putida* KT2440 the entire pathway is encoded in a single chromosomal operon around PP_2337 (KEGG genome annotation):

Locus	Gene	Enzyme	EC / KO
PP_2333	prpR	GntR-family transcriptional regulator	—
PP_2334	prpB	2-methylisocitrate lyase	4.1.3.30 / K03417
PP_2335	prpC	2-methylcitrate synthase	2.3.3.5 / K01659
PP_2336	**acnD**	2-methylcitrate dehydratase (2-methyl-*trans*-aconitate forming), Fe/S	4.2.1.117 / K20455
PP_2337	**prpF**	**2-methylaconitate isomerase**	**5.3.3.- / K09788**
PP_2338	prpD	2-methylcitrate dehydratase (cofactor-less)	4.2.1.79 / K01720
PP_2339	acnB	aconitate hydratase 2 / 2-methylisocitrate dehydratase	4.2.1.3 / 4.2.1.99 / K01682

PrpF (PP_2337) is **immediately adjacent to and co-operonic with acnD (PP_2336)** — the exact genomic pairing seen where the AcnD/PrpF partnership was experimentally proven. This is decisive functional-context evidence. The presence of the enzyme in this operon means PrpF is co-regulated with the propionate-catabolic machinery (2-MCC operons are typically induced by propionate/2-methylcitrate via the PrpR regulator).

Pathway flux with PrpF: propionyl-CoA → (PrpC) 2-methylcitrate → (AcnD) 2-methyl-*trans*-aconitate → **(PrpF) 2-methyl-*cis*-aconitate** → (AcnB aconitase) 2-methyl-D-isocitrate → (PrpB) pyruvate + succinate.

The physiological importance of this pathway in *Pseudomonas* is underscored by studies showing that disrupting 2-MCC enzymes (e.g., prpB) redirects propionate/propionyl-CoA into polyhydroxyalkanoate (PHA) synthesis, altering odd-chain monomer content (P 37995793) — evidence that the 2-MCC (of which PrpF is part) is the principal propionate-oxidation route in these bacteria.

5. Subcellular Localization

PrpF is a **soluble cytoplasmic enzyme**. Evidence: (i) all 2-MCC reactions occur in the cytosol where propionyl-CoA and TCA-cycle intermediates reside; (ii) the protein has no predicted signal peptide, no transmembrane segment, and no membrane/secretion annotation (UniProt keywords are only "Isomerase" and "Reference proteome"); (iii) the characterized orthologs were purified and assayed as soluble proteins from cytoplasmic extracts (P 18295882, P 14702315). It carries out its function in the cytoplasm, in direct metabolic coupling with the cytoplasmic dehydratase AcnD and aconitase AcnB.

6. Evidence Summary

Claim	Evidence type	Source
PrpF = 2-methylaconitate cis–trans isomerase	Direct in vitro enzymology (ortholog)	P 14702315 P 17567742 P 29145506
Isomerizes AcnD product → aconitase substrate	In vitro + NMR (ortholog)	P 29145506
Required with AcnD for propionate growth	In vivo complementation (*S. enterica*)	P 14702315
Cofactor-less; PLP-independent isomerase fold	X-ray crystallography (ortholog)	P 17567742 P 29145506
Catalytic residues Lys73, Cys107	Mutagenesis + structure; conserved in Q88KF3	P 29145506 + this analysis (83.9% id; K73/C107 conserved)
Membership in 2-MCC operon with acnD in *P. putida*	Genomic context (KEGG)	KEGG ppu:PP_2333–2339
Cytoplasmic localization	Sequence features + biochemical purification	UniProt Q88KF3; P 18295882
2-MCC is main propionate route in *Pseudomonas*	Mutant physiology	P 37995793

7. Supported and Refuted Hypotheses

- **Supported:** PrpF is a 2-methylaconitate isomerase acting in the AcnD-dependent 2-methylcitrate cycle (converts 2-methyl-*trans*- to 2-methyl-*cis*-aconitate). ✔

- **Supported:** PrpF is a cytoplasmic, cofactor-independent (non-PLP, non-metal) enzyme with catalytic Lys73/Cys107. ✔
- **Supported:** In *P. putida* KT2440, prpF is genomically embedded in a complete methylcitrate operon adjacent to acnD. ✔
- **Refuted:** PrpF is itself an aconitase/hydratase — no hydratase activity was detected; it is strictly an isomerase. ✘
- **Refuted (superseded model):** PrpF's role is merely to protect the AcnD Fe/S cluster from oxidative damage (an early hypothesis in P 14702315). Later enzymology (P 29145506) established the direct isomerase mechanism as the primary function.

8. Limitations & Future Directions

- Direct enzymatic characterization exists for the *S. oneidensis* ortholog, not for *P. putida* Q88KF3 specifically; the assignment relies on very high sequence identity (83.9%), conserved catalytic residues, and conserved operon context — strong but inferential.
- The precise identity of the AcnD product isomer (2-methyl-*trans*-aconitate vs. 4-methylaconitate) remains a point of mechanistic refinement in the literature.
- *P. putida* KT2440 encodes **both** PrpF/AcnD and a PrpD homolog (PP_2338); the relative in-vivo flux through each branch and any conditional regulation is not established and would benefit from targeted deletion + metabolomic studies.
- No experimental subcellular-localization or structural study of the *P. putida* protein has been reported; an AlphaFold model plus a *P. putida* prpF knockout growth phenotype on propionate would close the remaining gaps.